ENG: HUSSAM ZAMEL
NO: 0798508843

* EXPERIMENTAL PHYSICS *

* Significant figures and Errors *

← الأرقام المعنوية والأخطاء.

* المطلوب من التجربة:

* معرفة عدّ الأرقام المعنوية
* معرفة قواعد التقريب مع مراعاة الأرقام المعنوية.
* اجراء العمليات الحسابية مع مراعاة الأرقام المعنوية.
* معرفة الخطأ وايجاد الشك وايجاد الخطأ النسبي.

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* كيفية تحديد الأرقام المعنوية:

* جميع الأرقام الغير صفرية تعتبر أرقام معنوية

(أي 9-1 ارقام معنوية)

* 2.314 → 4 S.F ← (هنا أربع أرقام معنوية)
* 1.92 → 3 S.F (significant Figures) هنا تعني

* الأصفار الواقعة بين الأرقام المعنوية تعتبر أرقام معنوية.

* 2.06.705 → 6 S.F
* 7.203 → 4 S.F

* الأصفار الواقعة على يمين العدد وبعد الفاصلة العشرية تعتبر أرقام معنوية

* 2.00 → 3 S.F (فاصلة عشرية)
* 2.100 → 4 S.F

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* الأصفار الواقعة على يسار العدد وبعد الفاصلة العشرية لا تعبر أرقاماً معنوية.

* 0.00012 → 2 S.F

* 0.00 20130 → 5 S.F

* الأصفار الواقعة أما العدد الصحيح دائماً من الفواصل، يمكن اعتبارها جزء منها معنوي أو جبريها وذلك من خلال التعريب والقسمة.

* 700000 → $0.7 * 10^5$ → 1 S.F

→ $0.700 * 10^5$ → 3 S.F

* كيفية التقريب وقواعده

* اذا كان الرقم اكبر من 5 يعرب الرقم

* اذا كان الرقم اقل من 5 لا يعرب الرقم

* اذا كان الرقم 5 ويليه أرقام غير صفر يعرب

* اذا كان الرقم 5 ويليه صفر او لا شئ:

* انظر قبل اذا زوجي لا يعرب.

* انظر قبل اذا فردي يعرب

* قرب الأعداد التالية إلى أربع ارقام معنوية

* 97.5679 → 97.57 4 S.F

* 10.7349 → 10.73 4 S.F

* 239.556 → 239.6 4 S.F

* 7.65650 → 7.656 4 S.F

* 6.77755 → 6.778 4 S.F

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* اجراء العمليات الحسابية مع مراعاة الأرقام المعنوية:

* في حالة الجمع والطرح:

* في أي عملية حسابية يجب أن تكون عدد المنازل العشرية للإجابة بعدد المنازل العشرية الأقل من الأعداد الداخلة في العملية الحسابية.

* $3.862 + 1.3 = 5.162 = 5.2$

(3: عدد المنازل العشرية للعدد الأول — 1: هذا أقل منازل عشرية — 2: يجب أن يكون بعد الفاصلة رقم واحد حسب القاعدة)

* $7.23 + 5.071 + 1.5234 = 13.8244 = 13.82$

(2: الأقل منازل — 4: يجب أن يكون رقمين بعد الفاصلة حسب القاعدة)

* في حالة الضرب والقسمة:

* في الضرب والقسمة يجب أن عدد الأرقام المعنوية للإجابة النهائية هو أقل عدد أرقام معنوية الداخلة في العملية الحسابية.

* $7.6 * 1.74 = 13.224 \rightarrow 0.13224 * 10^2 \rightarrow 0.13 * 10^2$

(2: يجب أن يكون فقط رقمين معنويين)

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* $5.65 * 1.2 = 6.78 \rightarrow 6.8 \rightarrow 0.678 * 10^1 \rightarrow 0.68 * 10^1$

لازم أنه يكون رقمين

* $\frac{59106}{5} = 11821.2 \rightarrow 0.118212 * 10^5 \rightarrow 0.1 * 10^5$

لازم أنه يكون رقم واحد

* $6000 \div 6 = 1000 \rightarrow 1 * 10^3 \rightarrow 0.1 * 10^4$

لازم أنه يكون رقم واحد

---

* معرفة الخطأ وايجاد الشئ وايجاد الخطأ المئوي:

* الخطأ:

* جهاز ديجتال: ← الخطأ يكون أقل قراءة ΔL

* جهاز يدوي: ← الخطأ يكون أقل قراءة مقسوم على 2 $\frac{\Delta L}{2}$

* الشئ uncertainty ← $\Delta L, \Delta V, \Delta M, \ldots$

* الدقة في القياس ← $\frac{\Delta L}{L}, \frac{\Delta V}{V}, \frac{\Delta M}{M}$

* ايجاد الخطأ في حالة الجمع والطرح:

$C = A + B$ $\quad C = A - B$ ← الناقص لا يؤثر في ايجاد الخطأ

$\Delta C = \Delta A + \Delta B$ $\quad \Delta C = \Delta A + \Delta B$

ديجتال	يدوي
5.48 ← $\Delta = 0.01$	$\frac{0.01}{2} = 0.005$
7.5132 ← $\Delta = 0.0001$	$\frac{0.0001}{2} = 0.00005$

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* إيجاد الخطأ في حالة الضرب والقسمة:

$C = A * B$ $\qquad$ $C = \frac{A}{B}$

$\frac{\Delta C}{C} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$ $\qquad$ $\frac{\Delta C}{C} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$

* في حالة الأس:

$C = A^n * B^m$ $\qquad$ $C = A^n \pm B^m$

$\frac{\Delta C}{C} = n\frac{\Delta A}{A} + m\frac{\Delta B}{B}$ $\qquad$ $\Delta C = n\Delta A + m\Delta B$

* لو أريد الخطأ النتيجة حجم الكرة:

$V = \frac{4}{3}\pi r^3 \longrightarrow v = r^3 \rightarrow V = r * r * r$

الثوابت تهمل عند إيجاد الخطأ

$\frac{\Delta V}{V} = 3\frac{\Delta r}{r}$ $\qquad$ $\frac{\Delta V}{V} = \frac{\Delta r}{r} + \frac{\Delta r}{r} + \frac{\Delta r}{r}$

* لو أريد الخطأ النتيجة الكثافة

$\rho = \frac{M}{V} \longrightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta M}{M} + \frac{\Delta V}{V}$

---

* ملاحظة: كلما زادت المنازل العشرية كلما زاد الدقة

26.0 mm ② $\qquad$ 26.00 mm ① $\qquad$ 26. mm ③

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* الخطأ المئوي: (percenting error)

$$P.E = \frac{|(tru - exp)|}{tru} * 100\%$$

g = 9.81
g = 10

أسئلة على التجربة

1- Using significant rules to Find 37600/4 = 9400 → $0.9400 * 10^4$ ($0.9 * 10^4$)

2- If the relation between A & B given (C = A - B), if A = 5.28 and B = 1.8 then (C ± ΔC) is: C = A - B → C = 5.28 − 1.8 = 3.48 → 3.5

$$\Delta C = \Delta A + \Delta B$$

$\Delta C = 0.01/2 + 0.1/2$ → ΔC = 0.005 + 0.05 = 0.055 → 0.06

3- The number (5.000025) contains (7) significant figures. (T)

4- Express the results of the following calculations to the appropriate number of significant Figures?

a) 0.42 * 0.09 = 0.037 → 0.04

b) $\frac{1.36 * 0.51}{2.6} = 0.266769 \rightarrow 0.27$

c) 3.76 + 14.83 + 2.11 = 20.70 $2.07 * 10^1$

d) 37.76 + 3.907 + 226.4 = 268.067 → 268.1

5- If X ± ΔX = (4.00 ± 0.02) and Z = √X

Calculate Z ± ΔZ =

Z = √X Z = √4.00 = 2.00

$\Delta Z = X^{\frac{1}{2}}$ $\frac{\Delta Z}{Z} = \frac{1}{2}\frac{\Delta X}{X}$ → $\Delta Z = \frac{1}{2}\left(\frac{0.02}{4.00}\right) * 2.00 = 0.005$

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NO: 0798508843

6. a) $7.939 + 6.26 + 11.1 = 25.299 \rightarrow 25.3$

b) $(27.2 * 15.63)/1.846 = 230.30119 \rightarrow 230$ or $0.230 = 30119 * 10^{3}$

c) (0.07284) to 3 S.F $\rightarrow 0.0728$ $\quad 0.230 * 10^{3}$

d) (726.835) to 3 S.F $\rightarrow 727$

7. Determine the number of significant figures in following number:

a) $005.00 * 10^{-4}$ cm $\rightarrow$ 3 S.F b) 0.0350 gram $\rightarrow$ 3 S.F

c) 1.0003 joul $\rightarrow$ 5 S.F d) 50000 N $\rightarrow$ 5 S.F

8. if $X \pm \Delta X = (16.00 \pm 0.04)$ and $Z = \sqrt{X}$ calculate $Z \pm \Delta Z$.

9. Experimental value of gravity is found to be $10.07\ m/s^2$.
Accepted value "g" is $9.81\ m/s^2$. calculate the percent error.

$g = 10.07$ $\quad p.e = \left|\frac{g_{tru} - g_{exp}}{g_{tru}}\right| * 100\% \rightarrow \left|\frac{9.81 - 10.07}{9.81}\right| * 100$

$g_{tru} = 9.81$ $\quad p.e = 2.66\%$

10- A student wanted to calculate the density (ρ) of Mass (M).
if $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1)\ cm^3$
the absolute error of density ($\Delta\rho$) in: g/cm^3 is:

$\rho = \frac{M}{V} = \frac{10}{5} = 2$ $\quad \frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + \frac{\Delta V}{V}$

$\Delta\rho = \left(\frac{0.01}{10} + \frac{0.1}{5}\right) * 2 = 0.042 \Rightarrow 0.04$

11- The uncertainty for the area of circle, is $\Delta A = \left(\frac{2\Delta r}{r}\right) A \cdot (\pi)$

$A = \pi r^2 \rightarrow \frac{\Delta A}{A} = \frac{2\Delta r}{r}$ $\quad \Delta A = \left(\frac{2\Delta r}{r}\right) * A$

12- The number of significant Figures in the measurment $L = 75.00$ m $\rightarrow$ 4 S.F

13- The addition of the mesurment (6.201cm, 7.4cm, 0.68cm, 12.0cm) equal:

$6.201 + 7.4 + 0.68 + 12.0 = 26.281 \rightarrow 26.3$

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ENG: HUSSAM ZAMEL
NO: 0798508843

EXPERIMENTAL PHYSICS.

* Measurments.

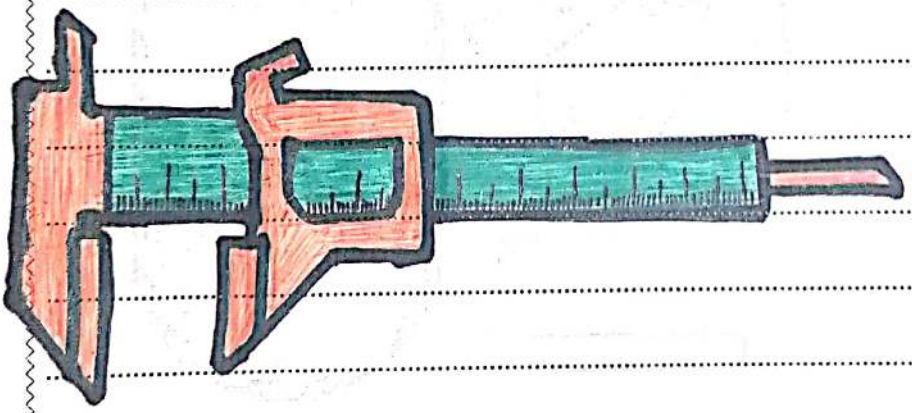
أجهزة القياس.

* الهدف من التجربة:
* معرفة أدوات القياس
* معرفة استخدامها
* معرفة نسبة الخطأ.

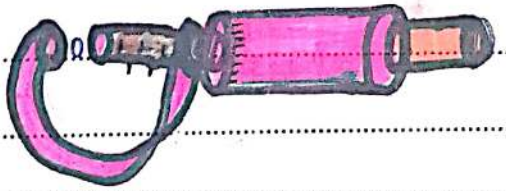
* معرفة أدوات القياسية:



* المسطرة (Ruler)



* الورنية (caliper)



* الميكرومتر (Micrometer)

ENG: HUSSAM ZAMEL
NO: 0798508843

EXPERIMENTAL PHYSICS.

* Measurments.

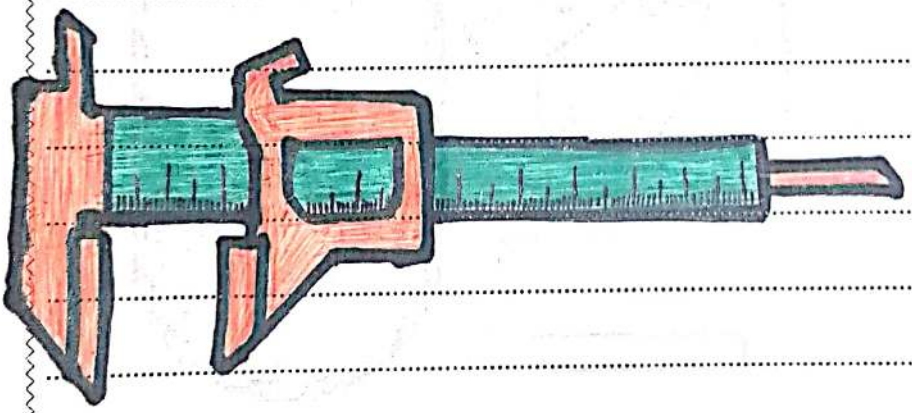
أجهزة القياس.

* الهدف من التجربة:
* معرفة أدوات القياس
* معرفة استخدامها
* معرفة نسبة الخطأ.

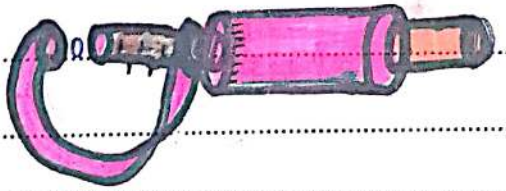
* معرفة أدوات القياسية:



* المسطرة (Ruler)



* الورنية (caliper)



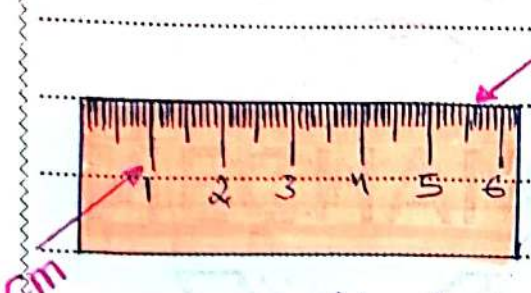
* الميكروميتر (Micrometer)

ENG: HUSSAM ZAMEL
NO: 0798508843

* معرفة استخداماتها

* معرفة نسبة خطأها

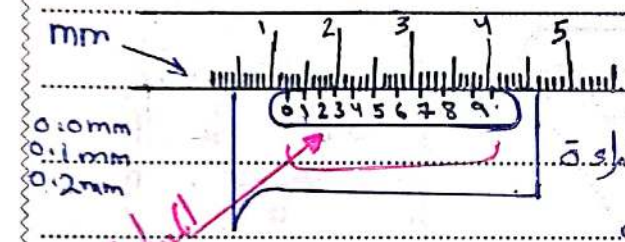
* المسطرة (Ruler):



- تتكون من تدريجين ← 1.0 mm
- اقل شيء تقيسه المسطرة 1mm ← 1.2 mm
- * نسبة الخطأ في المسطرة $\Delta L = \frac{1mm}{2}$ ← 1.3 mm
- نسبة الخطأ الثابتة في المسطرة ← $\Delta L = 0.5mm$

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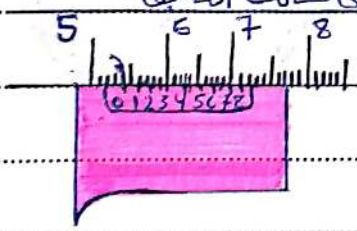
* الورنية (caliper):



- تتكون من مسطرة مثبتة تحدد القراءة ← 54.0mm
- اقل شيء تقيسه الورنية 0.1mm ← 41.2mm
- * نسبة الخطأ في الورنية $\Delta L = \frac{0.1mm}{2}$ ← 53.3mm
- نسبة الخطأ الثابتة في الورنية ← $\Delta L = 0.05mm$

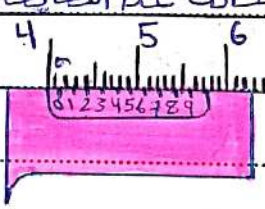
• كيفية قراءة الورنية:

(في حالة التطابق)



54mm + 0.0mm = 54mm

(في حالة عدم التطابق)

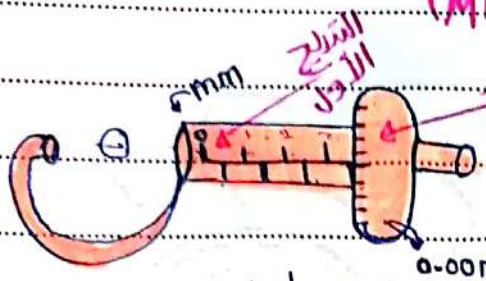


41-42

41mm + 0.2mm = 41.2mm

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* الميكرومتر: (Micrometer)



يتكون الميكرومتر من تدريجين

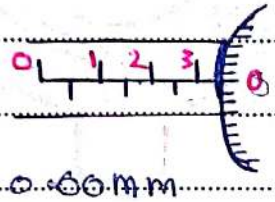
أقل تدريج يقيسه الميكرومتر 0.01mm

* نسبة الخطأ في الميكرومتر $\Delta d = \frac{0.01mm}{2}$

نسبة الخطأ في الميكرومتر ثابتة ← $\Delta d = 0.005$

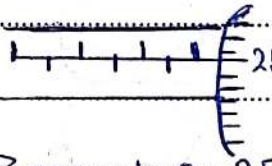
0.00mm
0.01mm
0.02mm

* كيفية قراءة الميكرومتر:



3mm + 0.00mm

3mm ∧ 3.00mm



0.50mm

3mm + 0.25mm = 3.25mm

* كل دورة كاملة في الميكرومتر ← 0.5mm

---

* اكتساب المهارة في التدريب العملي:

cm → mm * 10

* 35 cm → 35*10 → 350 mm — مسطرة
* 17.1 mm → ورنية
* 5.23 mm → ميكرومتر
* 19.0 mm → مسطرة 19 تختلف 19.0 ← ورنية
* 17. mm → مسطرة

* يجب أن تكون جميع الأرقام بـ (mm) لتتمكن من إجراء التحويلات

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* الاسطوانة

$A = 2\pi r L$ $\quad V = \pi r^2 L$

$\frac{\Delta A}{A} = \frac{\Delta r}{r} + \frac{\Delta L}{L}$ $\quad \frac{\Delta V}{V} = 2\frac{\Delta r}{r} + \frac{\Delta L}{L}$



* الاسطوانة البلاستيكية

$A_p = A_{ex} - A_{in}$ $\quad V_p = V_{ex} - V_{in}$

$A_p = 2\pi r_1 L - 2\pi r_2 L$ $\quad V_p = \pi r_1^2 L - \pi r_2^2 L$



$r = \frac{d}{2}$

$\Delta r = \frac{\Delta d}{2}$

* الكرة:

$A = 4\pi r^2$ $\quad V = \frac{4}{3}\pi r^3$ $\quad r = \frac{d}{2}$

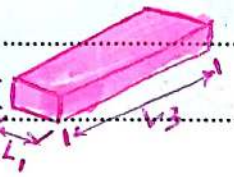
$\frac{\Delta A}{A} = 2\frac{\Delta r}{r}$ $\quad \frac{\Delta V}{V} = 3\frac{\Delta r}{r}$

$\Delta l = \frac{\Delta d}{2}$

* الخشبة:

$A = L_1 * L_2$ $\quad V = L_1 * L_2 * L_3$

$\frac{\Delta A}{A} = \frac{\Delta L_1}{L_1} + \frac{\Delta L_2}{L_2}$ $\quad \frac{\Delta V}{V} = \frac{\Delta L_1}{L_1} + \frac{\Delta L_2}{L_2} + \frac{\Delta L_3}{L_3}$



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NO: 0798508843

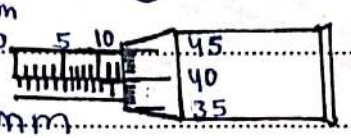
أسئلة على التجربة

1- Whic of the Following measurment represent the reading of micrometer:

a) 10.3 mm b) 8.0 cm c) 1.50 cm (d) 2.32 mm

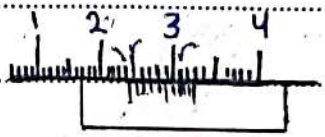
80 mm 15 mm

2- The reading of micrometer:



12 mm + 0.40 mm = 12.40 mm

3- The reading of the vernier caliper is:



24 mm + 0.7 mm = 24.7 mm

4- if we turned a closed micrometer 11 turns and $\frac{1}{5}$ turns then the micrometer reding is: $(11+\frac{1}{5}) * 0.5\,mm = 5.60\,mm$

28.7 mm

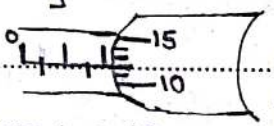
5- L = 2.87 cm $\Delta L = 0.05$ cm



$\Delta L = 0.5$ mm

mm → cm ÷10

6- If the diameter for the ball was mesured using a micrometer as Shown the $\Delta V = (0.075 * 10^{-9}\,m^3)$



$mm^3 \rightarrow$ *10^{-9}

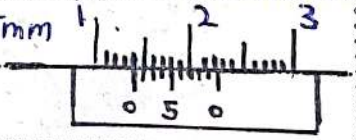
$V = \frac{4}{3}\pi r^3$ $r = 2.12/2 = 1.06$ mm → $V = 4.988\,mm^3$ $d = 2\,mm + 0.12\,mm$

$\frac{\Delta V}{V} = 3\frac{\Delta r}{r}$ $\Delta V = 3 * \left(\frac{0.005}{1.06}\right) * 4.988 = 0.075\,mm^3$ $d = 2.12$ mm $r = \frac{d}{2}$

7- The instrument used to take this reding (0.548 cm) is: micrometer

8- The reding of caliper showen is: 11.50 mm

5.48 mm



9- Which of the Following measurment micrometer:

(a) 9.25 mm b) 8.0 cm c) 9.60 cm d) 2 mm e) 10.1 mm

80

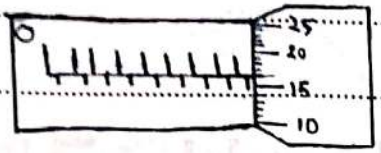
10- A vernier caliper was used to measured of cube what is the error in length ΔL is: a) 0.50 mm (b) 0.05 mm c) 0.5 mm d) 0.005 m

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11- if a student used ruler a maximum range (10 cm) to measure the length of a table whic has the length (1.65 m) then the ΔL:

ΔL = 0.5mm ✗ m → cm ×100 1.65 m → 165 cm → $\frac{165}{10}$ = 16.5 = 17

17 * 0.5mm = 8.5mm

12- 

8.5mm + 0.16mm = 8.66 mm

13- Ahmad measured the radius and the mass of sphere using micrometer and the balance R = 0.86 mm and m = 4.3g calculate the voulme (V ± ΔV) and density (ρ ± Δρ)

$V = \frac{4}{3}\pi r^3$ ← r = 0.86 → V = 2.664 mm^3 → 2.7 mm^3

$\frac{\Delta V}{V} = 3\frac{\Delta r}{r}$ ← $\Delta V = 3 * \left(\frac{0.01/2}{0.86}\right) * 2.664 mm^3 = 0.046 mm^3$ ≈ $0.05 mm^3$ (0.005)

$\rho = \frac{M}{V} = \frac{4.3}{2.7} = 1.6\ g/mm^3$

$\frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + \frac{\Delta V}{V}$ → $\Delta\rho = \left(\frac{0.1/2}{4.3} + \frac{0.05}{2.7}\right) 1.6 = 0.06\ g/mm^3$

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EXPERIMNTAL PHYSICS

* Vector :

المتجهات :

* المطلوب من هذه التجربة :

* كيفية إيجاد المحصلة استخدام أمايلي :

* التحليل

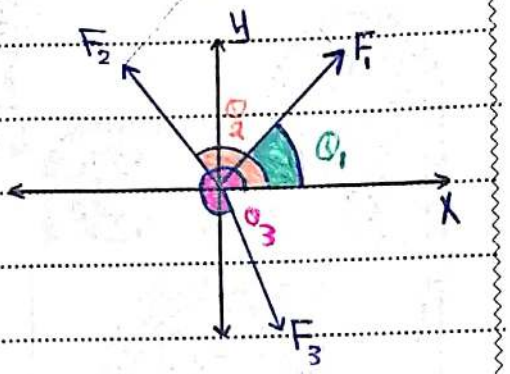
* الجمع المتجهي

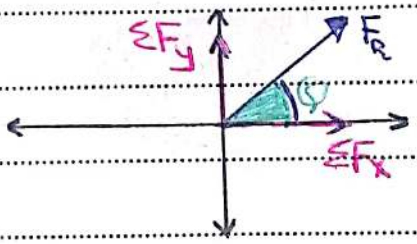
* قانون ← sin law و cos law

---

* التحليل :

FX (axis)	Fy (axis)
$F_1 \cos\theta_1$	$F_1 \sin\theta_1$
$F_2 \cos\theta_2$	$F_2 \sin\theta_2$
$F_3 \cos\theta_3$	$F_3 \sin\theta_3$
ΣF_x	ΣF_y





$$F_R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

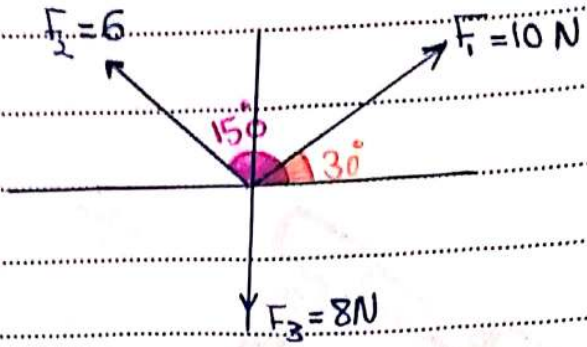
$$\theta = \tan^{-1} \frac{\Sigma F_y}{\Sigma F_x}$$

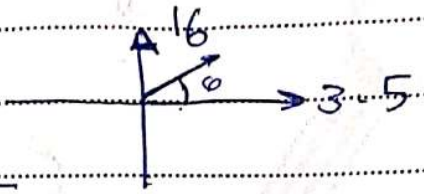
* عندما يكون الجسم متزن يجب $\Sigma F_x = 0$ و $\Sigma F_y = 0$

* محصلة متجهين في حالة الاتزان تساوي مقدار المتجه الثالث ولكن عكس الاتجاه

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ΣF_x	ΣF_y
$F_1 \cos\theta_1$	$F_1 \sin\theta_1$
$10\cos 30 = 8.7$	$10\sin 30 = 5$
$F_2 \cos\theta_2$	$F_2 \sin\theta_2$
$6\cos 150 = -5.2$	$6\sin 150 = 3$
$F_3 = 0$	$8\,N$
3.5	16





$F_R = \sqrt{16^2 + 3.5^2} =$

$\theta = \tan^{-1} \frac{16}{3.5} =$

---

* الجمع المتجهي.

عبارة عن تمثيل القوى عن طريق الرسم مع تحديد مقياس رسم مناسب ومن خلالها نوجد المحصلة

$F_1 = 3N$	$F_2 = 4N$	$F_3 = 4.3N$
$\theta_1 = 30$	$\theta_2 = 150N$	$\theta_3 = 250$

1N ← 1cm

4.3N ← 4.3cm

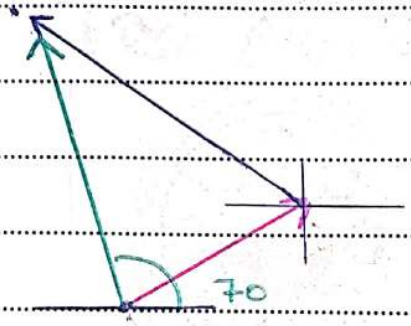
F_R

4.3 cm

$4.3 = F_R$

$\theta_3 = 70 + 180$

$\theta_3 = 250$



* دائماً المحصلة يكون ذيلها مع ذيل الأول ورأسها مع رأس الأخير

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* قانون Sin law و Cos law

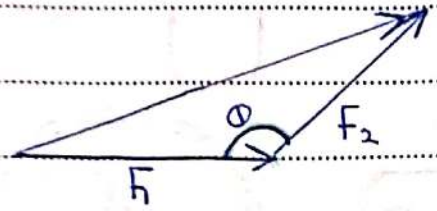
* $\text{Cos law} = F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$ *

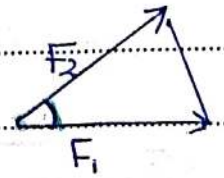
بين قوتين متجهين

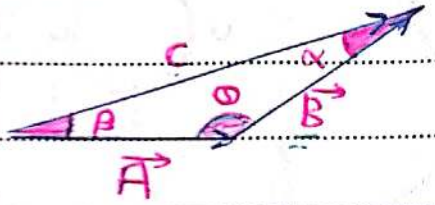
* $\text{Cos law} = F_{net} = \sqrt{F_1^2 + F_2^2 - 2F_1F_2\cos\theta}$

بين قوتين الزاوية بينهما منفرجة

* $\text{Sin law} = \frac{\sin\beta}{B} = \frac{\sin\alpha}{A} = \frac{\sin\theta}{C}$



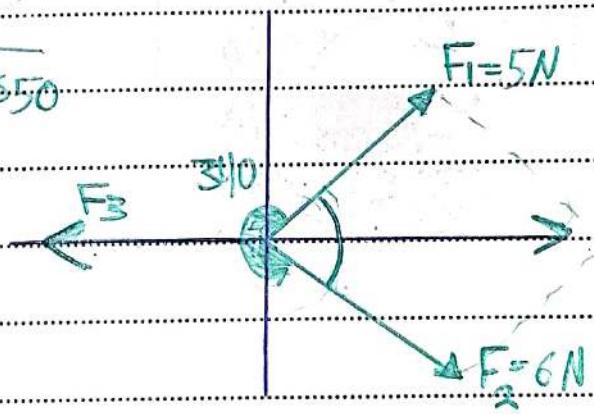




* هذا القانون لا يطبق إلا على متجهين

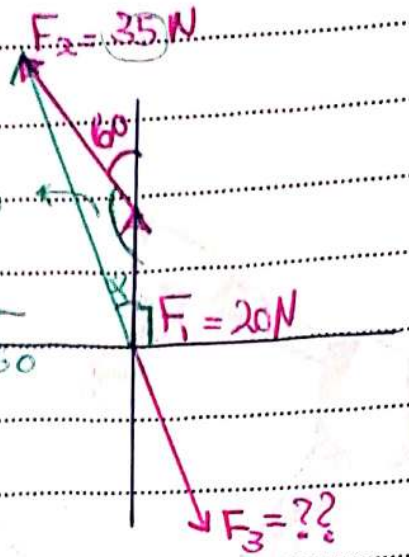
$|F_{net}| = |F_3| = \sqrt{5^2 + 6^2 + 2*5*6\cos 50}$

$|F_{net}| = |F_3| = 9.97$



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Cos Law

$180 - 60 = 120$

$|F_{net}| = |F_3| = \sqrt{20^2 + 35^2 + 2(20)(35)\cos 60}$

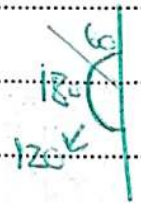
$F_{net} = F_3 = 48.2\ N$

$F = 48.2$

Sin Law



$\frac{\sin\alpha}{35} = \frac{\sin 120}{48.2} =$



$\sin\alpha = 0.6$

$\alpha = \sin^{-1} 0.6$

$\alpha = 38.9$

$\theta_{net} = 38.9 + 90 = 128.9$

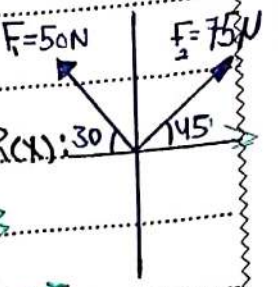
$\theta_3 = \theta_{net} + 180$

$\theta_3 = 308.9^\circ$

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اسئلة من التجربة

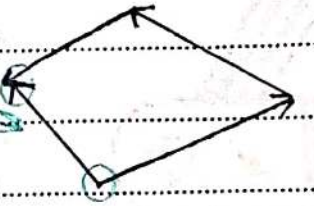


1- The horizontal component of resultant Force R(x):

$F_{2x} = 75 \cos 45 \longrightarrow 53$ $R_x = 53 - 43.3$

$F_{1x} = 50 \cos 30 \longleftarrow 43.3$ $\therefore R_x = 9.73 \longrightarrow$

2- if the scale used to draw this vector diagram for three force was (1 mm = 1N) then the magnitude of the net force is :



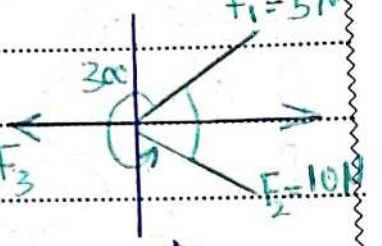
1 mm ⟶ 1N

20 mm ⟶ F $\quad F = 20N$

3- F_1, F_2, F_3 are three forces in equilibrium. if you know that ($F_{3x} = 1.32N$, $F_{3y} = -7.5N$) the direction of resultant between (F_1 and F_2) according to X- positive axis is :

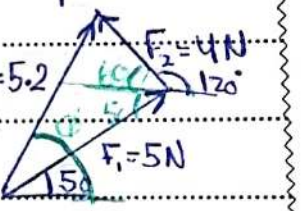
$$\theta = \tan^{-1} \frac{-7.5}{1.32} = -80 + 180 = 100$$

4- For equilibrium state, the magntude F_3 is:



$\sqrt{5^2 + 5^2 + 2 \times 5 \times 5 \times \cos 60} = 13.22N$

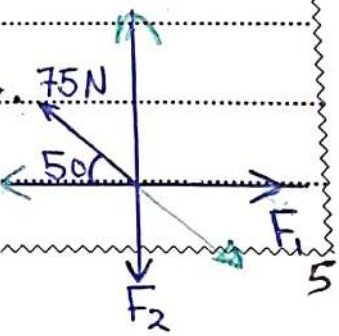
5- The direction of (R) according to X positive:



$$\frac{\sin\theta}{4} = \frac{\sin 110}{5.2} = 46.28 + 50 = 96.28$$

6- A stationary object has three force acting on it. 75N

what are the magntude F_1 is:

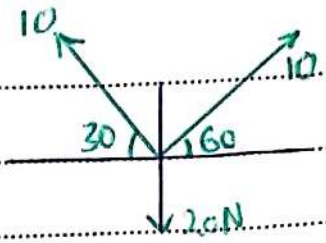


$F_1 = 75 \cos 50 = +48.2$

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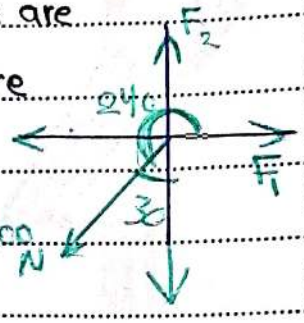
7- in the Figure shown, what is the magnitude of the resultant force in (N) along X axis:



$10\cos 60 = 5$ ⟵ ⟶

$10\cos 30 = 8.66$ $\quad 8.66 - 5 = 3.66$ ←

8- The set of force shown in the following figures are equilibriume, the magnitudes of F_1 and F_2 in (N) are respectively given by:



$F_1 = 200\cos 240 = 100$

$F_2 = 200\sin 240 = 173$

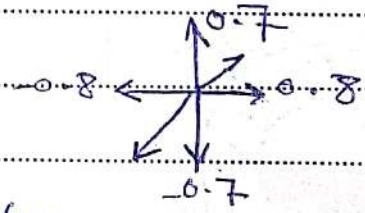
9-

mass	force	direction	X	Y
80g	0.8	0	0.8	0
70g	0.7	90	0	0.7
106g	1.06	41.2	+0.8	+0.7

$F = mg$ $\quad g \to kg \div 1000$

$F_1 = \frac{80}{1000} * 10 = 0.8$

$F_2 = \frac{70}{1000} * 10 = 0.7$



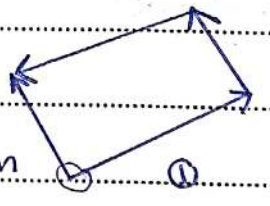
$F_3 = \sqrt{0.8^2 + 0.7^2} = 1.06$

$\theta = \tan^{-1}\frac{0.7}{0.8} = 41.2$

$m = \frac{F}{g} = \frac{1.06}{10} = 0.106\ kg \to 106g$

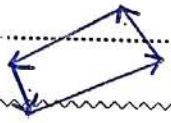
10- Find the resultant force:

if (0.5mm = 1N)



$0.5\ mm \longrightarrow 1N$ $\quad \frac{0.5}{0.5}F = \frac{1.5}{0.5}$

$1.5\ mm \longrightarrow F$ $\quad F = 30N$



F = zero

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11-

$F = mg$

Vector	mass(kg)	Force(N)	Direction	X
A	0.2	2	30	1.7
B	0.2	2	120	-1
C	0.28	2.8	75.5	+0.7

$F_1 = 0.2 * 10 = 2$

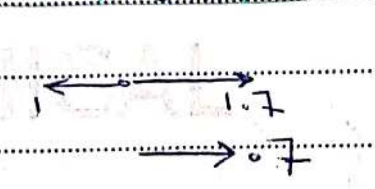
$F_2 = 0.2 * 10 = 2$

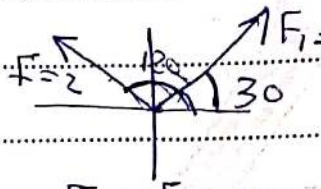
$2\cos 30 = 1.7$

$2\cos 120 = -1$

$\sqrt{2^2+2^2+2*2*2*\cos 90} = 2.8$

$m = \frac{F}{g} \quad \frac{2.8}{10}$



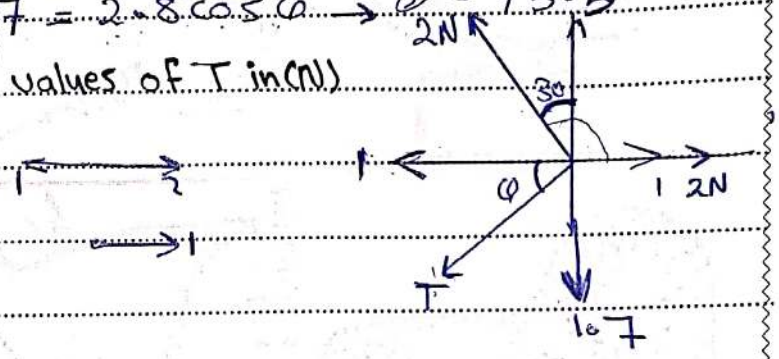


$F_x = F\cos\theta$

$+0.7 = 2.8\cos\theta \rightarrow \theta = 75.5$

12- For equilibruim , The values of T in(N) and θ.

$F_y = 2\cos 30 = 1.7$



$T = \sqrt{1.7^2+1^2} = 1.97$

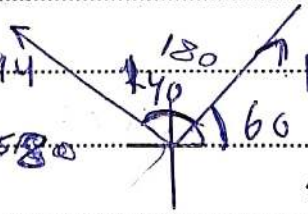
$\theta = \tan^{-1}\frac{1.7}{1} = 59.53$

13- Find the resultant force of the following tow force:

$F_1 = 1.96$ (wieght 200gram) at 60° From positive X axis

$F_2 = 2.94$ (wieght 300gram) at 140 From positive X axis

The resultant force in(N) and dirction.





$\sqrt{1.96^2+2.94^2+2*1.96*2.94\cos 80} = 3.81$

$\frac{\sin\alpha}{2.94} = \frac{\sin 80}{3.81} \quad \alpha = 49.5 + 60$

$\theta = 109.5$

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EXPERIMNTAL PHYSICS.

* Newton's second law:

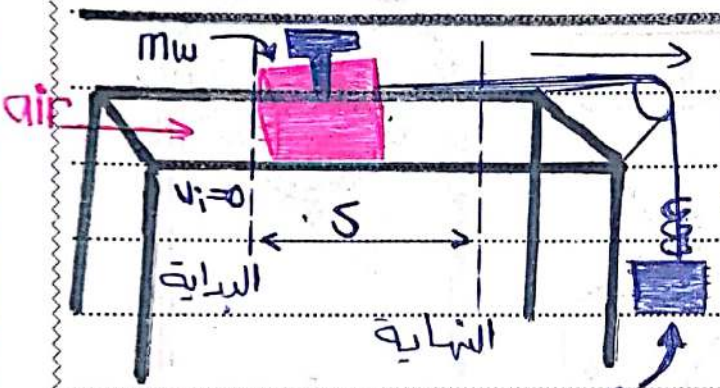
$$F = ma$$

قانون نيوتن الثاني

* الهدف من هذه التجربة.

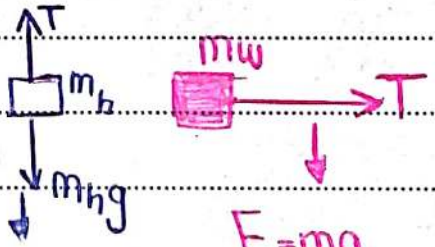
* معرفة القانون الثاني

والممارسة في التجريبية.



(الكتلة المعلقة) (hanging mass)

* الآن نريد أن نشتق العلاقة:



$F = m_h a$

$F = ma$

$T = m_w a \quad \ldots ②$

$\ldots ① \quad m_h g - T = m_h a$

$$m_h g = (m_w + m_h)\, a$$

$$S = \cancel{V_i t}^{0} + \frac{1}{2} a t^2 \rightarrow S = \frac{1}{2} a t^2 \rightarrow a = \frac{2S}{t^2} \rightarrow S \rightarrow$$ تعطى في السؤال

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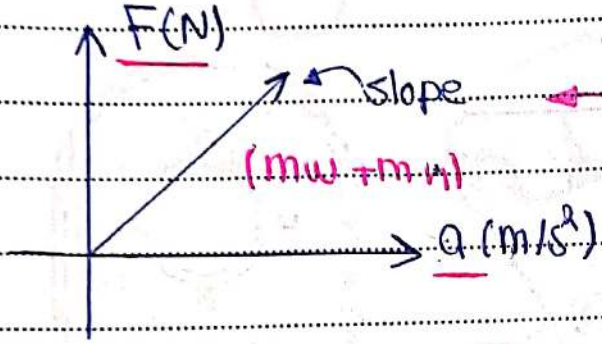
* $m_h g = (m_w + m_h) a$
* $m_h g = (m_w + m_h) \frac{25}{t^2}$

$F = m_t \cdot a$

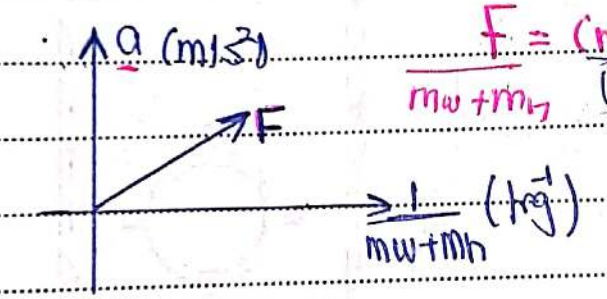
$F = m a$

* $\vec{F} \propto \vec{a}$ ← when the tota mass is constant
* $a \propto \frac{1}{m}$ ← when the Force is constant

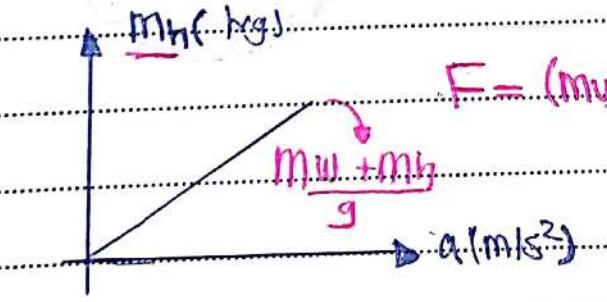
* $y = \text{slope}\ x$



$F = (m_w + m_h) a$



$F = (m_w + m_h) a \rightarrow a = F \frac{1}{(m_w + m_h)}$

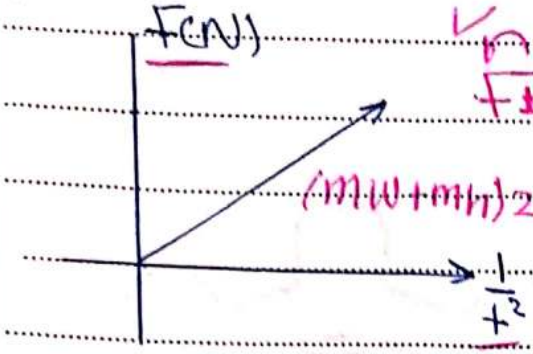


$F = (m_w + m_h) a \rightarrow m_h g = (m_w + m_h) a$

$m_h = \frac{m_w + m_h}{g} a$

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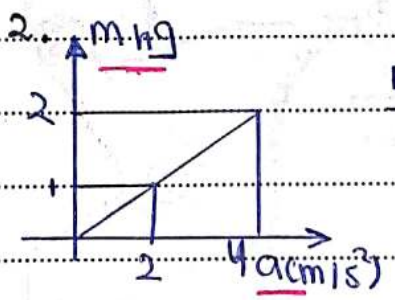
$F = (m_w + m_h)\,a \rightarrow F = (m_w + m_h)\frac{2s}{t^2}$

$(m_w + m_h)2s$

$F = (m_w + m_h)2s\,\frac{1}{t^2}$

اسئلة على التجربة

1- Expermental setup shown in Fig1 when $m_a + m_h$ were kept contant and equal 210g, the graph between $m_h g$ the hanging mass times gravity acceleration and a is accleration is hown in Fig 2.



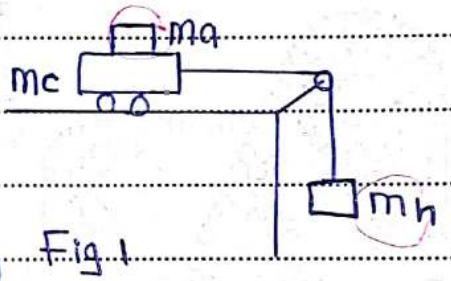


Fig 1

$m_h + m_a = 210$

$m_h + 120 = 210$

$m_h = 90\ g$

a) The mass of cart mc in (g) is :

b) if the added mass is $m_a = 120$ g the magntude of acceleration for this system :

$m_a + m_h = 210$ g

$m_c = ??$

$F = (m_w + m_h)\,a$

$m_h g = (m_a + m_c + m_h)\,a$

$\text{Slope} = m_a + m_h + m_c = \frac{2-1}{4-2} = 0.5$ kg

$m_a + m_h + m_c = 500$g $\rightarrow 210 + m_c = 500$g $\quad m_c = 290$g

$F = (0.5)\,a \rightarrow m_h g \rightarrow 0.09 \times 10 = 0.9$

$0.9 = 0.5\,a \rightarrow a = 0.9/0.5 \rightarrow a = 1.8\ m/s^2$

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2- if $s = 0.8m$

a) Complete the following table :

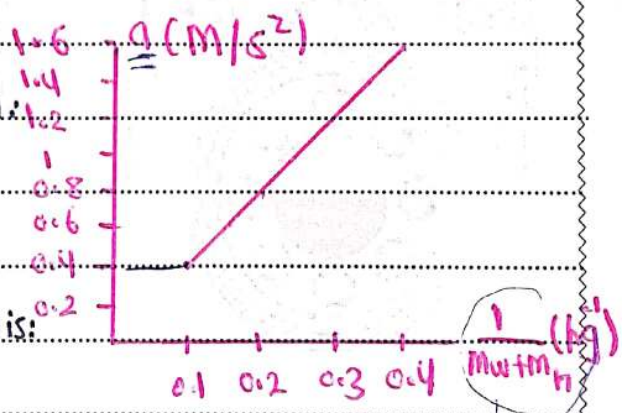
hanging mass (kg)	Time	Force (N)	acceleration (m/s^2)	Total mass (kg)
0.02	1.4	0.2	0.8	0.25
0.04	0.98	0.4	1.66	0.24
0.06	0.8	0.6	2.5	0.24

$F = m_h g = 0.02 * 10 = 0.2$

$a = \frac{2s}{t^2} \rightarrow \frac{2*0.8}{1.4^2} = 0.8$

$F = m_{tot} a$

3- the relation between the acceleration (a) and the total mass (m) withe constant Force is hown in Figurer.



a) The applied Force on the system:

$Slope = F = \frac{1.6 - 0.4}{0.4 - 0.1} = 4N$

b) The hanging mass m_h in gram is:

$F = m_h g$

$\frac{4}{10} = m_h = 0.4$

$\frac{F}{m_{tot}} = \frac{m_{tot}\, a}{m_{tot}}$

$a = F \frac{1}{m_{tot}}$

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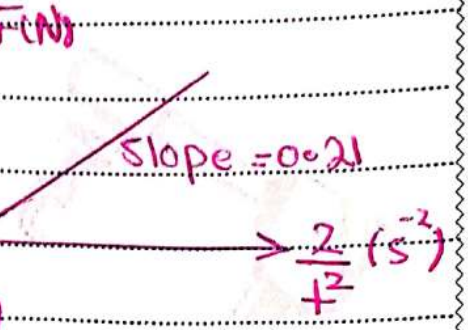
4- if $m_{tot} = 300g$ the distance which (mw) moved in all trials is: 5

$F = m_{tot}\, a$

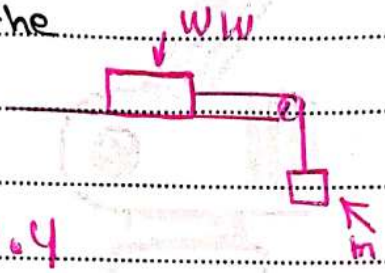
$F = m_{tot}\, \frac{2S}{t^2}$ → $F = m_{tot} S \frac{2}{t^2}$

slop $= m_{tot} S = 0.21$

$0.35 = 0.21$ $S = 70\,cm$



5- if $m_w = 160g$, $m = 40g$ then the acceleration is:



$F = m_{tot}\, a$ ← $F = m_h g$

$\frac{0.4}{0.2} = \frac{0.2}{0.2} a$ → $a = 2\,m/s^2$ $0.04 * 10 = 0.4$

6- the constant hanging mass in the experiment is:

(m_h)

$F = m_h g$

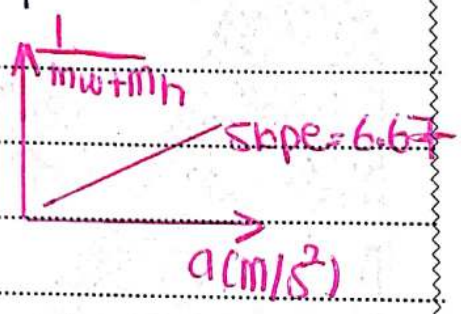
$\frac{0.15}{10} = m_h$

$0.015\,kg$ $15\,g$

$\frac{F}{m_{tot}} = \frac{m_{tot}\, a}{m_{tot}}$

$a = F \frac{1}{m_{tot}}$

$F = \frac{1}{slope} = 0.15$



7- in Netwons second low expremantal, you can calculate always v_f as $v_f = \frac{1}{t^2}$ (F) $v_f = v_i + at$

$v_f = \frac{2S}{t^2}$ $t = \frac{2S}{t}$

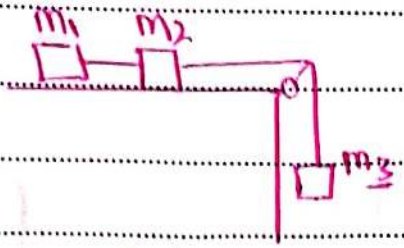
8- in Netwons second low expremantal the decline of air track acts on results. (T)

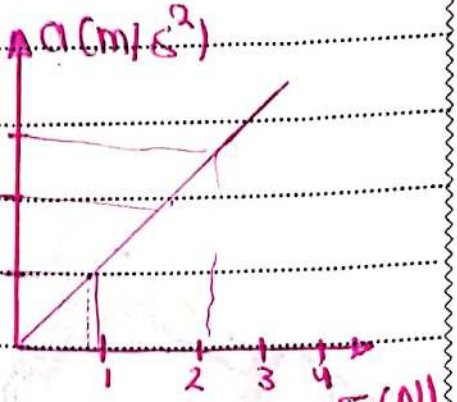
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lab PHYSICS 101

9-





$F = m_{tot}\, a$

$a = \frac{1}{m_{tot}} F$

a) what is the sum three mass.

$\frac{1}{m_{tot}} = \text{slope} = \frac{3-1}{2-1} = 2 = \frac{1}{m_{tot}}$ $\quad \frac{2}{2} m_{tot} = \frac{1}{2}$

$m_{tot} = 0.5\ kg$

b) Find Fnet whe $m_3 = 0.4\ kg$

$F = m_h g \rightarrow 0.4 \times 10 = 4\ N$

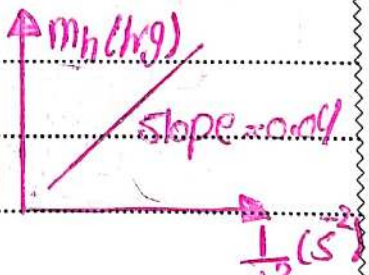
c) what is the acceleration of the system when

$m_3 = 0.3\ kg$

$F = m_{tot}\, a$ $\quad F = m_h g = 0.3 * 10 = 3$

$\frac{3}{0.5} = \frac{0.5}{0.5} a \rightarrow a = 6\ m/s^2$

7 b- if the distance that (mw) moved (70cm) and total mass (200g) the the acceleration gravity is:



$F = m_{tot}\, a$

$m_h g = \frac{m_{tot}}{g} \frac{2s}{t^2}$

$m_h = \frac{m_{tot}}{g} \frac{2s}{t^2}$

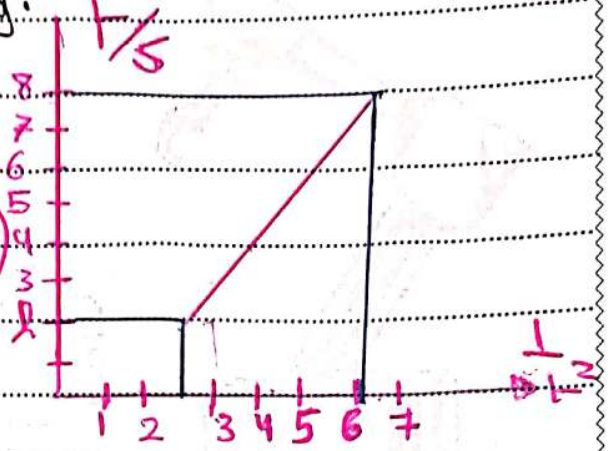
$\frac{m_{tot}\, 2s}{g} = 0.04$

$\frac{0.2 * 2 * 0.7}{g} = 0.04$

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8

11- the distance that Mw was moved in all trials considered to be (55cm) the the use the graph to answer the Following:



a) The physical meaning of the slope

$2 m_{tot} \rightarrow 2(m_w + m_h)$

b) The total mass

0.85 kg

c) If (t=.5 s.) the hanging mass (m_h) will be:

0.374 kg

$S = 55cm$

$F = m_{tot} a_1$

$\frac{F}{S} = m_{tot} \frac{2(S)}{t^2}$

$\frac{F}{S} = m_{tot} 2 \frac{1}{t^2}$

$Slope = \frac{8-2}{6-2.5} = 1.7$

$\frac{2 m_{tot}}{2} = \frac{1.7}{2}$

$m_{tot} = 0.85$

$F = m_h g$

$\frac{3.74}{10} = m_h$

$m_h = 0.374 kg$

$f = m_{tot} a \rightarrow a = \frac{2S}{t^2}$

$f = 0.85 * 4.4$

$f = 3.74 N$

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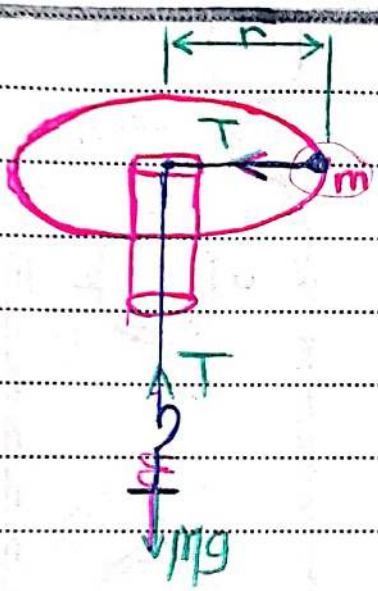
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Expermantal physics:

* Uniform circular motion:

الحركة الدائرية المنتظمة:

* المطلوب من هذه التجربة:
* معرفة العلاقة وكيفية التبديل فيها.



$a_c = \frac{v^2}{r}$, $v = \frac{x}{t} \rightarrow \frac{2\pi r}{T}$

$F_c = ma_c$

$T = ma_c$ but $T = mg$

$mg = m\frac{4\pi^2 r^2}{T^2 r}$

$F_c = Mg$

$Mg = \frac{4m\pi^2 r}{T^2}$

$\frac{1}{T} = f \rightarrow$ frequency التردد

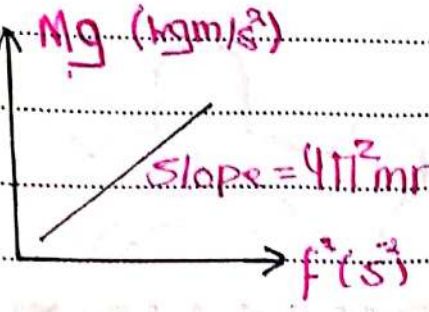
$$Mg = 4\pi^2 mrf^2$$

* M=? (Fixed r Change M).
* M=? (Fixed M Change r).

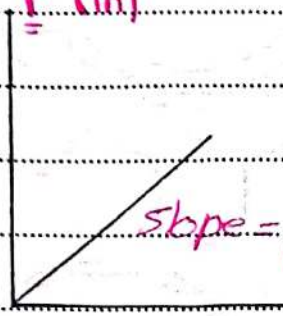
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$Mg = 4\pi^2 m r f^2$ $\quad y = \text{slope} \cdot x$



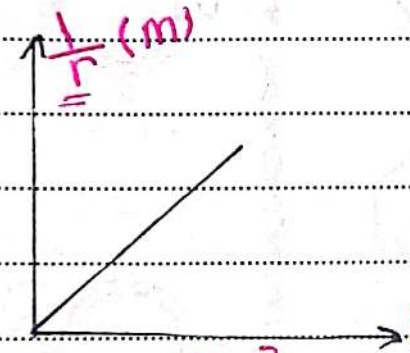
$Mg = 4\pi^2 m r f^2$



$Mg = 4\pi^2 m r f^2$

$\frac{Mg}{4\pi^2 m \frac{1}{T^2}} = \frac{4\pi^2 m r \frac{1}{T^2}}{4\pi^2 m \frac{1}{T^2}}$

$r = \frac{Mg}{4\pi^2 m} T^2 \qquad f^2 = \frac{1}{T^2}$



$\frac{Mg}{r} = 4\pi^2 m \frac{r}{r} f^2$

$\frac{1}{r} \frac{Mg}{Mg} = \frac{4\pi^2 m}{Mg} f^2$

$\text{slope} = \frac{4\pi^2 m}{Mg} \qquad \frac{1}{r} = \frac{4\pi^2 m}{Mg} f^2$

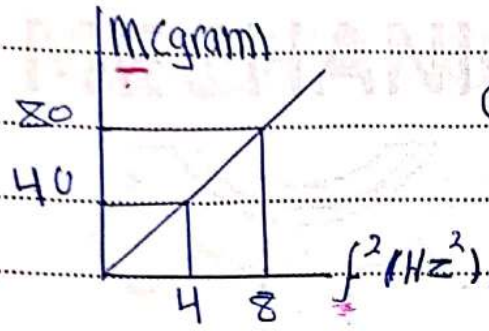
$P.E = \frac{|m_{th} - m_{ex}|}{m_{th}} * 100\%$

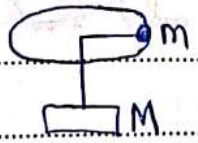
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اسئلة على التجربة

1- in the circular motion expermental, a student a studied the relation between the hanging mass (M) and the squre of the frequency (f^2) for relation body (m) using the radius (R=0.6m). A plotted data is shawen in the figure bellow:





$\frac{Mg}{S} = \frac{4\pi^2 r m f^2}{g}$

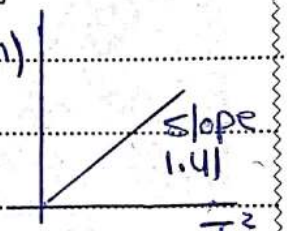
$M = \frac{4\pi^2 r m}{g} f^2$

$Slope = \frac{4\pi^2 r m}{g} = \frac{80-40}{8-4} = 10 \rightarrow \frac{g*10}{4\pi^2 r} = \frac{4\pi^2 m}{4\pi^2 r} = 4.22kg$

2- is uniform circulare motion, if (m=10g), (M=45g) then the acceleration of gravity is:

m=10g
M=45g
g

$\frac{Mg}{4\pi^2 m f^2} = \frac{4\pi^2 r m f^2}{4\pi^2 m f^2}$



$r = \frac{Mg}{4\pi^2 m f^2} \rightarrow r = \frac{Mg}{4\pi^2 m} T^2$ $T^2 = \frac{1}{f^2}$

$\frac{Mg}{4\pi^2 m} = 1.41 \rightarrow \frac{1.41*4*\pi^2*0.01}{0.045} = 12.36 m/s^2$

3- in uiform circulare motion experment, the raduis r=50cm M=40g, m=12g the velocity of m/s is:

r=50cm
M=40g
m=12g

$Mg = 4\pi^2 r m f^2$

$Mg = m a_c$

$Mg = m \frac{V^2}{r}$

$V = \frac{2\pi r}{T}$

$V^2 = \frac{4\pi^2 r^2}{T^2}$

$V = \sqrt{\frac{Mgr}{m}}$ $\sqrt{\frac{0.4*10*0.5}{0.012}} = 4.1 m/s$

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4- Aman is moved from rest around circalr path with radius 60m and he made 10 turns in 10 minute. this final velocity in (m/s) is:

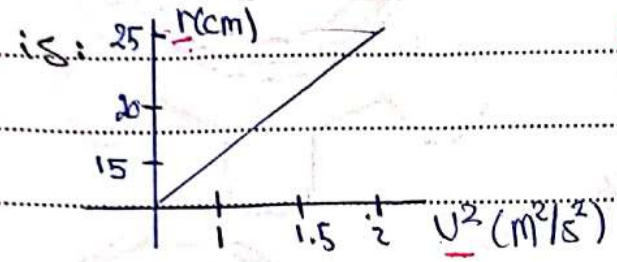
$r = 60m$

$10 \rightarrow 10 minut$

$= \rightarrow =$ $\rightarrow \frac{10T}{10} = \frac{10}{10} \rightarrow T = 1 minut \rightarrow 1*60 = 60$

$V = \frac{2\pi r}{T} = \frac{2\pi * 60}{60} = 6.28$ m/s

5- From the graph. The expermantal centripetal accelaration is:



$a_c = \frac{V^2}{r}$

$r \frac{a_c}{a_c} = \frac{V^2}{a_c}$

$Slope = a_c = \frac{20-15}{1.5-1} = (10 m/s^2)$ $r = a_c V^2$

6- in the circular motion expermantal, a student studied the relation between the centripetal force and the squre speed (V^2) for rotational body (m) by making the radius ($r = 0.45$ m).

a) Complete the tabel:

M (kg)	T (s)	F_c (N)	V (m/s)	$V^2 (m/s)^2$
0.02	1.15	0.2	2.45	6
0.04	0.82	0.4	3.44	11.83
0.06	0.67	0.6	4.22	17.8

$F = Mg = 0.02 * 10$

$F = 0.2$ N

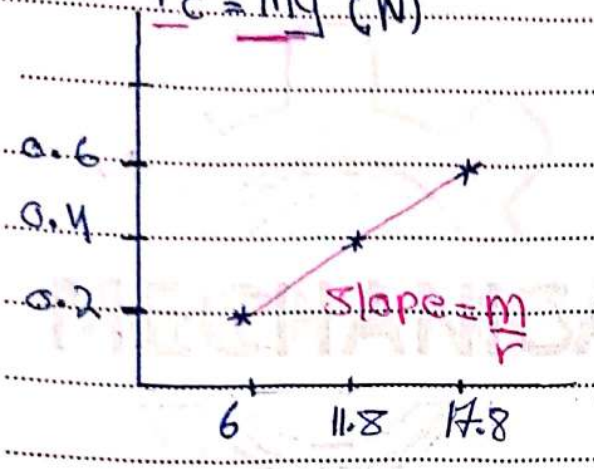
$V = \frac{2\pi r}{T} = \frac{2 * \pi * 0.45}{1.15}$

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b) Plot suitable graph between Fc and V^2

$F_c = Mg$ (N)



$Mg = 4\pi^2 r m f^2$

$Mg = mac \rightarrow Mg = m\frac{V^2}{r}$

$Mg = \frac{m}{r} V^2$

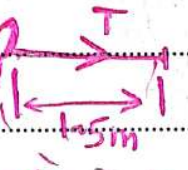
c) By using the slope of the graph, find the mass (m)

$Slope = \frac{m}{r} \rightarrow m = Slope * r = 0.45 * 0.03 =$

$m = 0.0156 kg \rightarrow 16g$

7- A ball of mass 0.5kg is attached to the end a string 1.5m long and whirled in horizontal plane aconstan speed if the cord can with stand a maximum tension of 50N what is the maximum speed of the ball befor the string breaks.

$T = 50N$ $\quad F_c = mac$ $\quad m = 0.5kg$



V $\quad T = m\frac{V^2}{r}$

$V = \sqrt{\frac{rT}{m}} = \sqrt{\frac{1.5m * 50}{0.5kg}} = 12.2 m/s$

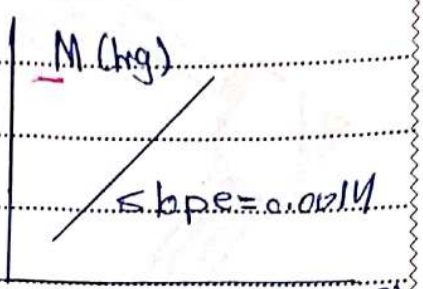
8- If m=10 gram, M=50 gram. what is the frequency in(Hz) to make the body m rotate in circular motion with the radius R=0.18m

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$\frac{Mg}{4\pi^2 rm} = \frac{4\pi^2 rmf^2}{4\pi^2 rm}$ $\quad f = \sqrt{\frac{Mg}{4\pi^2 rm}} = \sqrt{\frac{0.05*10}{4*\pi^2*0.18*0.01}} = 2.65\,Hz$

9- The mass which is rotate in uniform circular motion (m) in kg

$Mg = 4\pi^2 rmf^2$



slope $= \frac{m}{g}$

$Mg = mac$

$\frac{Mg}{g} = \frac{mac}{g}$

$0.0114*10 = m$

$m = 0.114 = 114g$ $\quad M = \frac{m}{g} ac$

10 - When $ac = 295.78\ m/s^2$ and $r = 30\ cm$ then the time of 20 revolutions is:

T ⟶ 20 revolutions

0.2 ⟵ 1 revolutions

$T = 0.2*20 = 4s$

$ac = \frac{v^2}{r}$

$ac = \frac{4\pi^2 r}{T^2}$

$ac = \frac{4\pi^2 r}{T^2}$

$T = \sqrt{\frac{4\pi^2 r}{ac}}$

$T = \sqrt{\frac{4*\pi^2*0.3}{295.78}}$

$T = 0.2$

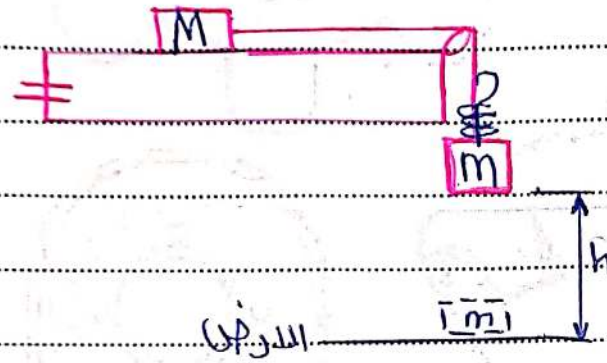
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Expremantal physics:

* Gravitional potential energy:

* مبدأ الطاقة
* الطاقة بالطريقة التجريبية:
* معرفة الجاذبية الأرضية.



$\Delta K + \Delta U = 0$

$K_F + U_F = K_i + U_i$

$\frac{1}{2}(M+m)V_F^2 + mgh_F = \frac{1}{2}(M+m)V_i^2 + mgh_i$

$U_i = K_F$

$\frac{1}{2}(M+m)V_F^2 = mgh_i$ $\quad V_F = \frac{L}{T}$

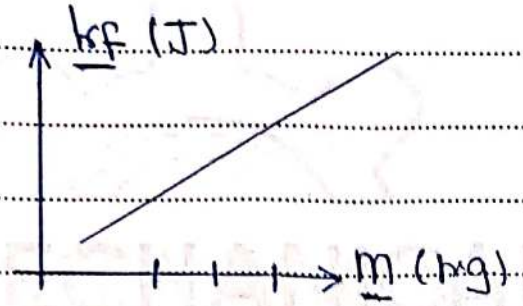
$K_F = mgh$

* g (h constant & Chang m).
* g (m constanant & Chang h).

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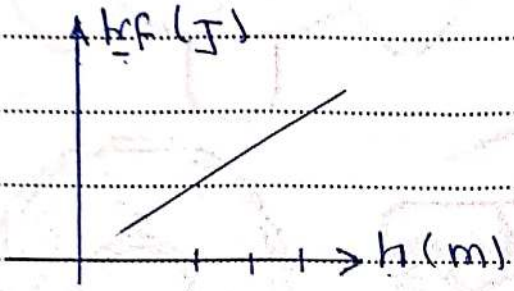
$K_f = mgh$



$K_f = mgh$

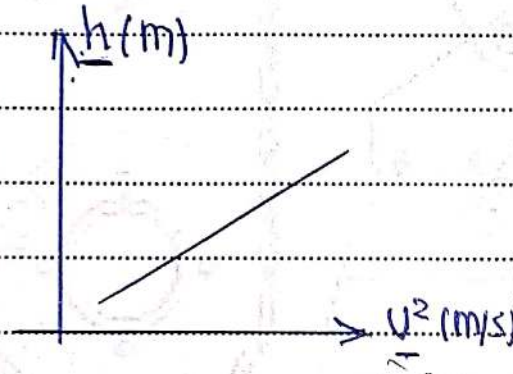
$K_f = ghm$

$Slope = gh$



$K_f = mgh$

$Slope = mg$



$K_f = mgh$

$\frac{1}{2}\left(\frac{M+m}{mg}\right)v^2 = \frac{mgh}{mg}$

$h = \frac{M+m}{2mg} v^2$

$Slop = \frac{M+m}{2mg}$

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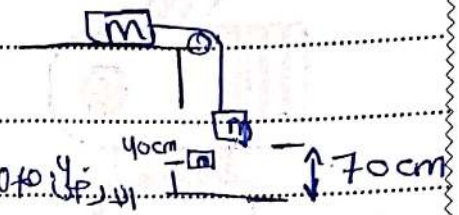
* اسئلة من التجربة

1- In gravitional potential energy expriment, before start move toward down from (hi=70 cm) the mass was (m=20g) and (M=200g), then the kinetic energy when (m) reach (h=40 cm) is:

$\frac{1}{2}mv^2$

$U_i = 0 \quad K_i = 0 \quad K_f + U_f = K_i + U_i$

$v = ? \quad K_f = ? \quad K_f = U_i - U_f$

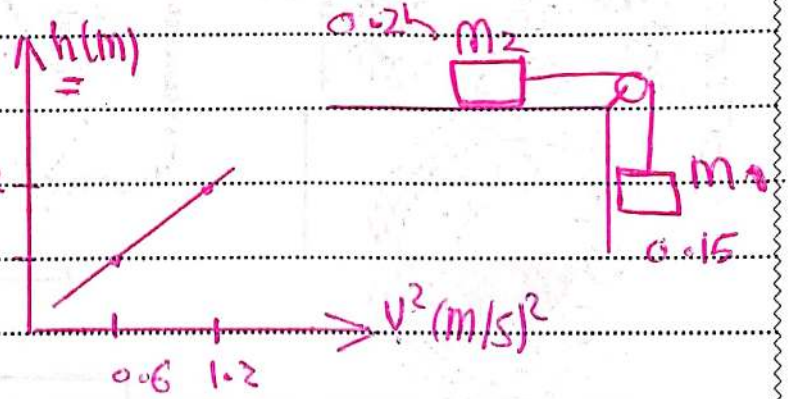


$K_f = mgh_i - mgh_f \rightarrow 0.02 \times 10 \times 0.7 - 0.02 \times 10 \times 0.4$

$K_f = 0.06\ J$

2- In conservation of energy expirement, a student obtained data plotted in the following figures masses $m_1 = 0.15 kg$ and $m_2 = 0.2 kg$ The acceleration gravity ~~decrease~~ :

$g = ??$



$Slope = \frac{m_2 + m_1}{2m_1 g}$

$\frac{0.2 - 0.1}{1.2 - 0.6} = 0.1667$

$K_f = m_1 g h$

$0.1667 = \frac{0.35}{0.3g}$

$\frac{1}{2}(m_2 + m_1)V^2 = m_1 g h$

$\frac{(m_2 + m_1)}{mg} \quad \frac{m_1 g h}{mg}$

$0.1667 \times 0.3g = 0.35$

$h = \frac{m_2 + m_1}{2mg} V^2$

$g = 7 m/s^2$

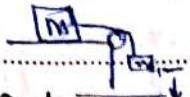
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3- in gravitational potential energy experiment, the chang in potential energy (ΔU) just for (m). (F)

$\Delta U = mgh$

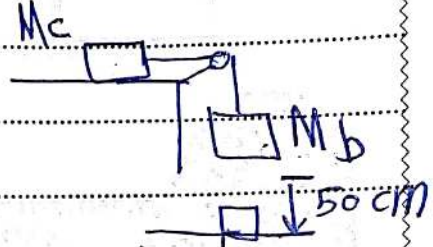
4- in gravitional potential energy experiment. when (m) reach the ground, the speed of (M) will increase. (T)



5- A hanging $M_b = 0.5$ kg and the mass of the cart $M_c = 1$kg The system is released from rest. the height of the haning mass is $h = 50$cm

a) complete the following table

U_i(J)	U_f(J)	K_i(J)	K_f(J)
2.5	0	0	2.5



b) the speed of the cart ____ m/s at the haning mass just hits the ground.

$U_i = 0$ U_f K_i

$U_i = mgh_i$ $U_f = mgh_f$ $\frac{1}{2}(M_c + M_b)v_i^2$

$0.5 * 10 * 0.5 = U_f = 0$

2.5J $U_f = 0$

$U_i = K_f$

$K_f = \frac{1}{2}(M_b + M_c)V_f^2 \rightarrow \sqrt{\frac{2K_f}{M_b + M_c}} = V \rightarrow 1.8$ m/s

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Expermantal phsics:

* Collisions :

* التصادمات :

* الهدف من هذه التجربة معرفة حساب الزخم وتوعيته :

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* أنواع التصادمات :

* التصادم المرن ← Elastic collisions

* التصادم الغير المرن ← Inelastic collisions

* كيف يمكنني معرفة نوع التصادم :

* من خلال الطاقة الحركية اذا :

← الطاقة محفوظة ← Elastic collision

← الطاقة غير محفوظة ← Inelastic collision

* prove that the Linear momentum always conserved *

يرهن أن الزخم دائماً محفوظ *

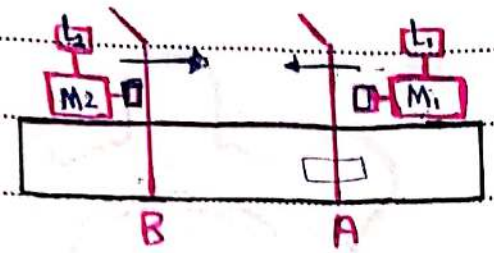
* prove that K.E is conserved in elastic collisions *
* and Unconserved in inelastic collisions *

يرهن أن الطاقة الحركية محفوظة في التصادم المرن ،
وايضا غير محفوظة في التصادم الغير مرن .

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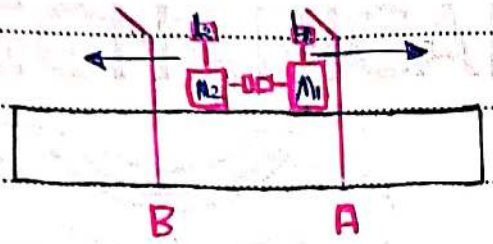
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* Elastic Collision:



الجسم 1 — A_1 — $V_{1i} = \frac{-L_1}{A_1}$

الجسم 2 — B_1 — $V_{2i} = \frac{+L_2}{B_1}$



A_2 — $V_{1f} = \frac{+L_1}{A_2}$

B_2 — $V_{2f} = \frac{-L_2}{B_2}$

* $P = MV$

$P_i = P_f$

$\rightarrow M_1V_{1i} + M_2V_{2i} = M_1V_{1F} + M_2V_{2F}$

* $K.E = \frac{1}{2}mV^2$

$K.E_i = K.E_F$

$\rightarrow \frac{1}{2}m_1V_{1i}^2 + \frac{1}{2}m_2V_{2i}^2 = \frac{1}{2}m_1u_{1F}^2 + \frac{1}{2}m_2u_{2F}^2$

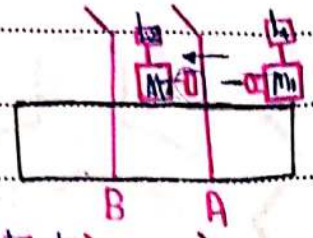
* الزخم كمية متجهة (الإشارة تهمنا).

* الطاقة محفوظة في المصدر الأول.

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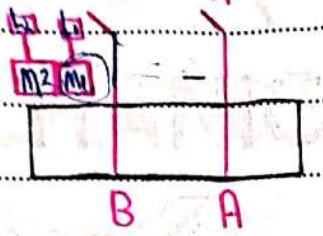
*Inelastic collisions:



الجسم 1 — الجسم 2

A_1 — 0

$V_{1i} = -\frac{L_1}{A_1}$ — $V_{2i} = 0$



الجسم 1 — الجسم 2

B_2 — B_1

$V_F = \frac{L_2}{B_1}$ or $\frac{L_1}{B_2}$

* $V_F = \frac{L}{\frac{B_1+B_2}{2}}$

* $P = MV$

$P_i = P_F$

$m_1 V_{1i} + m_2 V_{2i} = (m_1+m_2) V_F \rightarrow m_1 V_{1i} = (m_1+m_2) V_F$

* $K_i = K_F$

$\frac{1}{2} m_1 V_{1i}^2 + \frac{1}{2} m_2 V_{2i}^2 = \frac{1}{2}(m_1+m_2) V_F^2$

* الزخم دائماً محفوظ.

* الطاقة غير محفوظة في التصادم الغير مرن.

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أسئلة على التجربة

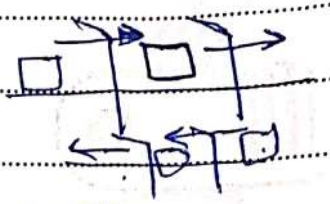
1- The energy is always conserved in all collision (F).

2- in elastic collision $K.E_f = K.E_i$. (T)

3- in an inelastic one dimension collision ($U_{1i} = 5 m/s$), ($U_f = 2 m/s$), and ($U_{2i} = 0$), ($M_2 = 6g$) then (M_1) is :

$m_1 v_{1i} = (m_1 + m_2) v_f$

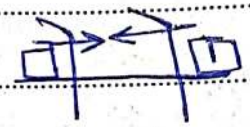
$m_1 (+5) = (m_1 + 6g)(+2)$



$5m_1 = 2m_1 + 12g \quad 3m_1 = 12g \quad m_1 = 4g$

4- in one dimension an elastic collision ($m_1 = 2m_2$) ($U_{1i} = 3 m/s$), ($U_{2i} = 5 m/s$) and linear momentum after collision ($P_f = 8 kg.m/s$) The M_1 :

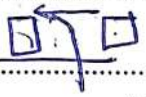
$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$



$2m_2 3 + m_2 5 = 8 kg \quad 8$

$6m_2 + m_2 5 = 8 kg \rightarrow m_2 = 8 kg \quad m = 2m_2 = 16 kg$

5- in the collision between carts all of the velocity was transferred from the first cart to the second cart what type of collision: inelastic collis



6- you are going to give the cart1 a push and collide it with cart 2, which is initially at rest. After the collision cart1 goes in the opposite direction. use the following data to complete the table for cart1 and cart 2 before and after the collisions.

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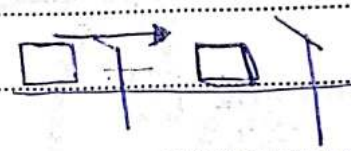
* The mass $m_1 = 100\ g$
* The mass $m_2 = 200\ g$
* The Distance $L = 4\ cm$

$t_{1i} = 0.165\ s$ $t_{1f} = 0.485\ s$ $t_{2f} = 0.246\ s$

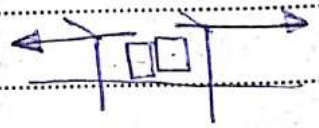
	Before coll	after coll
The velocity of cart 1 V_1(m/s)	+0.24	-0.082
The velocity of cart 2 V_2(m/s)	0	+0.163
The total momentum P(kg·m/s)	0.024	0.024
The total K.E (J)	$2.88 * 10^{-3}$	$2.99 * 10^{-3}$

* Is the total momentum conserved ? 0.024
* is the collision elastic or inelastic? inelastic

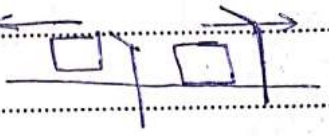
$$V_1 = \frac{L}{t} = \frac{0.04}{0.165} = 0.24\ m/s$$



$$V_{1f} = \frac{0.04}{0.485} = 0.082\ m/s$$



$$V_{2f} = \frac{0.04}{0.246}$$



$$P_{befor} = m_1 V_1 + m_2 V_2$$
$$0.1 * 0.24 + 0.2 * 0$$
$$0.024$$

$$P_{after} = m_1 V_1 + m_2 V_2$$
$$0.1 * (-0.082) + 0.2 * 0.163$$
$$P = 0.024$$

$$K = \frac{1}{2} m_1 V_1^2 + \frac{1}{2} m_2 V_2^2 = 0.5 * 0.1 * (0.24)^2 = 2.88 * 10^{-3}$$

$$K = 0.5 * 0.1 * (-0.082)^2 + 0.2 * 0.5 * (0.163)^2 = 2.99 * 10^{-3}$$

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